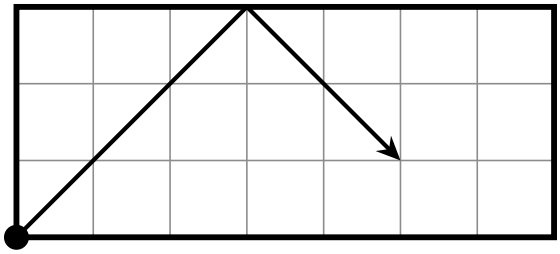
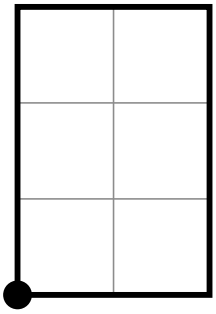
The ball starts at the dot in the bottom left corner of a table and rolls up and to the right, crossing each square from corner to corner. It bounces off every wall it hits and stops when it reaches a corner of the table. Each time the ball hits a wall counts as one bounce; reaching the corner at the end does not count. A 7 by 3 table is 7 squares wide and 3 squares high.



7 by 3

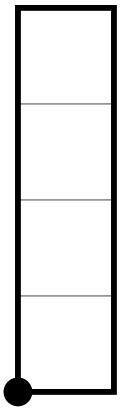
Partners take turns: one predicts where the ball will stop, and the other traces the path.

Problem 1: Trace the path on each table. Under each table, write the corner where the ball stops and the number of bounces.

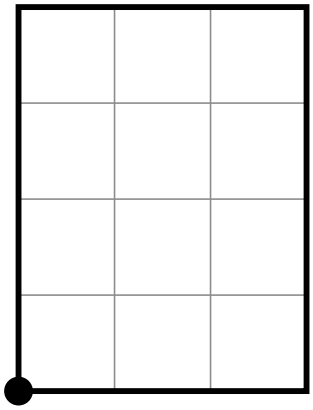


2 by 3

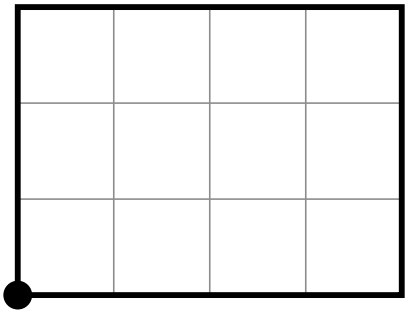
3 by 2



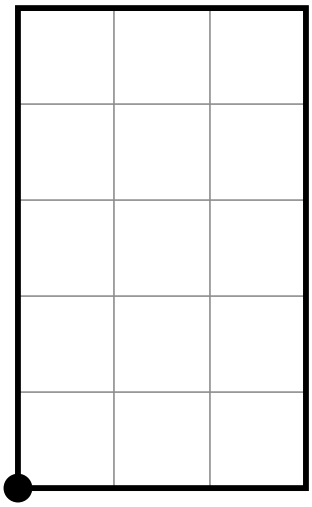
1 by 4



3 by 4



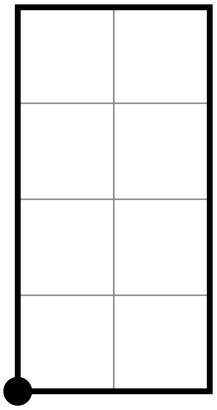
4 by 3



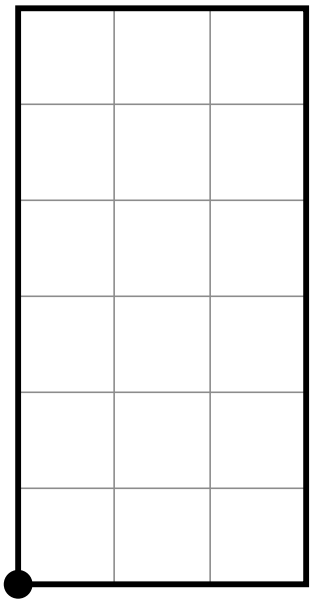
3 by 5

Problem 2: Trace the path on each table, and write the corner where the ball stops and the number of bounces.

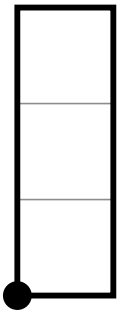
1 by 2



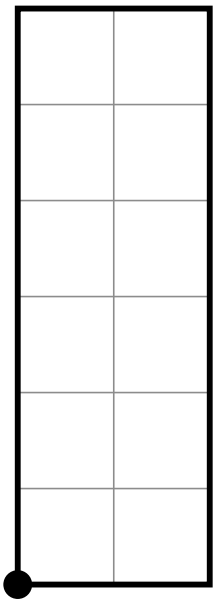
2 by 4



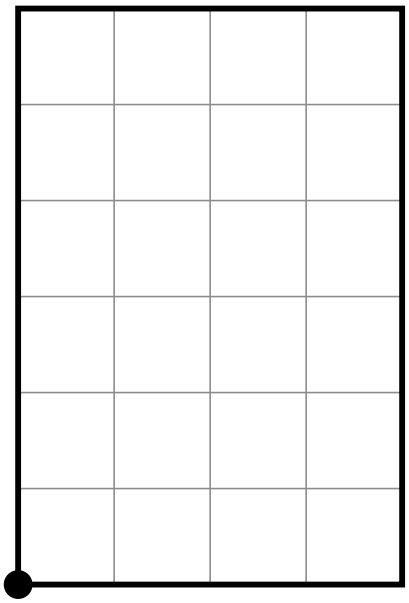
3 by 6



1 by 3

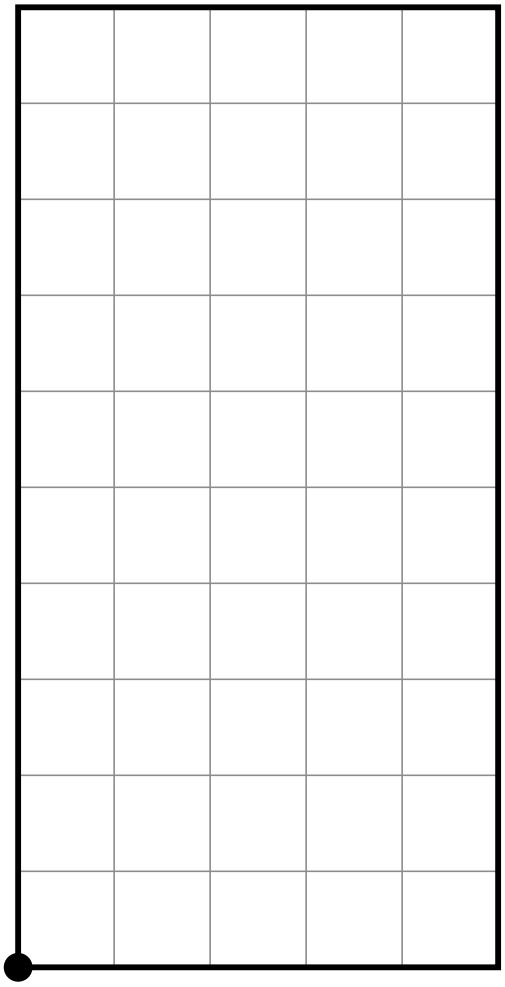


2 by 6

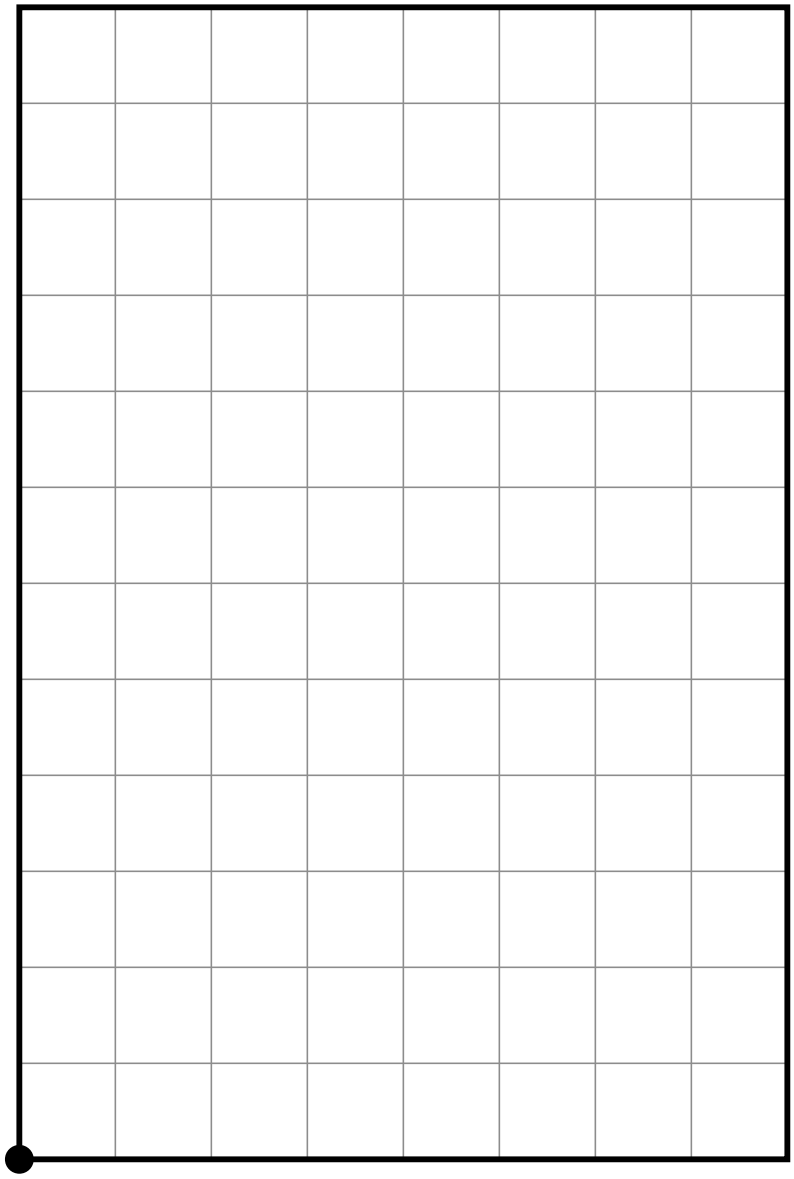


4 by 6

Problem 3: Before you trace these two tables, write the corner where you think the ball will stop and the number of bounces you expect. Then trace the paths to check.

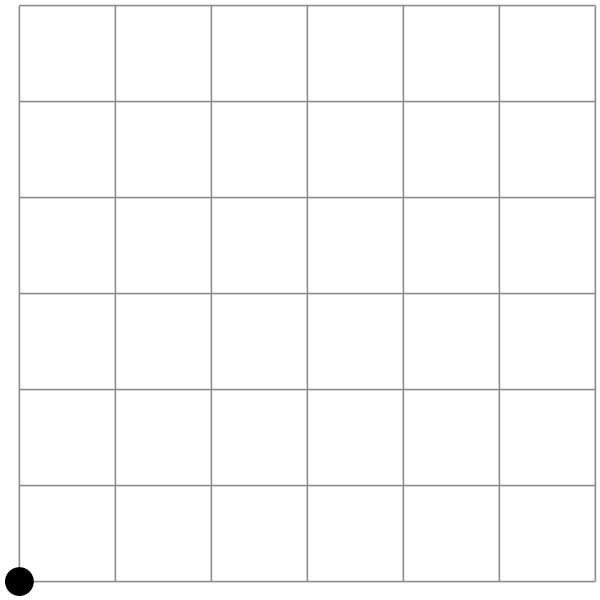


5 by 10

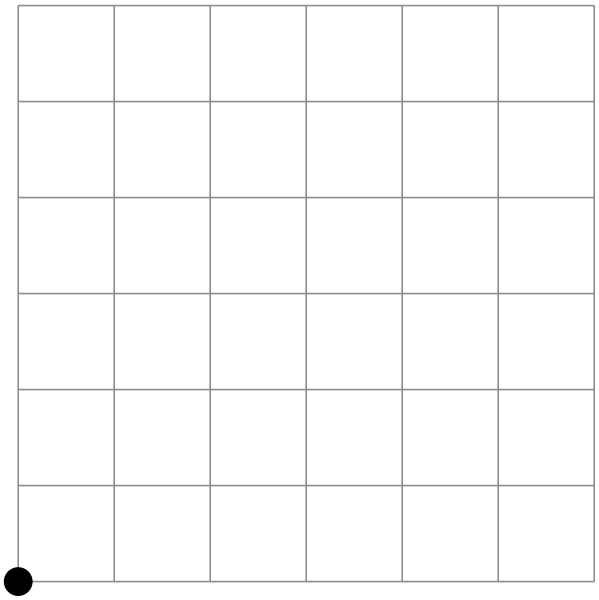


8 by 12

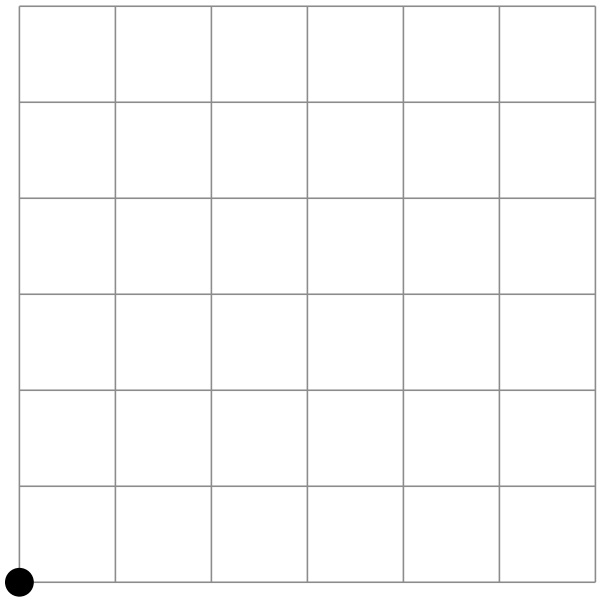
Problem 4: On each grid, draw a table with its bottom left corner at the dot, so that the ball does what is written under the grid. If there is no such table, explain why.



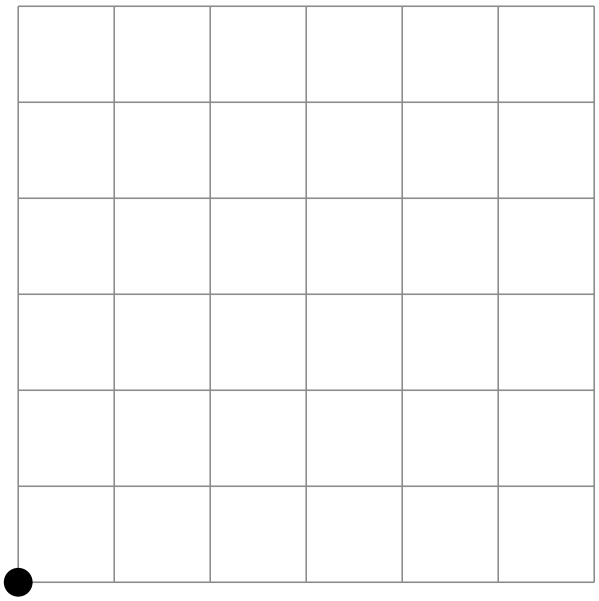
stops at the bottom right
after 1 bounce



stops at the top right
after 4 bounces



stops at the top left
after 7 bounces



stops at the top right
after 3 bounces